Baire Category and the Three Fundamental Theorems

Undergraduate project — working notes

Working draft

These notes develop familiar consequences of completeness. Most arguments are written out; three proofs are still drafts, and three established results have no proof here yet. The final conjecture is a classical exercise left unresolved in this notebook, not a claim of a new open problem. Background: elementary metric topology, sequences, and linear algebra.

1 Baire category

Definition 1.1 (Complete metric spaces)**.** For a metric space (X, d), write

$$B(x, r) = \{y \in X : d(x, y) < r\}, \qquad \overline{B}(x, r) = \{y \in X : d(x, y) \le r\}.$$

The second notation denotes the closed ball; it need not be the closure of the open ball. A sequence (x_n) is *Cauchy* if, for every $\varepsilon > 0$, there is N such that $d(x_m, x_n) < \varepsilon$ for $m, n \ge N$. The space is *complete* if every Cauchy sequence converges in X. A set is *dense* if it meets every nonempty open subset of X.

Definition 1.2 (Nowhere dense, meagre, and residual)**.** In a metric space as in Definition 1.1, a subset A is *nowhere dense* if $\operatorname{int}(\overline{A}) = \varnothing$. It is *meagre* if it is a countable union of nowhere dense sets. It is *residual* if $X \setminus A$ is meagre. All closures and interiors are taken in the ambient space X. A *Baire space* is a space in which every countable intersection of open dense sets is dense.

Lemma 1.3 (Nested closed balls)**.** *Let (X, d) be complete in the sense of Definition 1.1. Suppose $K_n = \overline{B}(x_n, r_n)$ are nonempty closed balls, $K_{n+1} \subseteq K_n$, and $r_n \to 0$. Then $\bigcap_{n=1}^{\infty} K_n$ contains exactly one point.*

Proof. For $m \ge n$, the centre x_m belongs to K_n, so $d(x_m, x_n) \le r_n$. Thus (x_n) is Cauchy and has a limit $x \in X$. For each fixed n, all later centres lie in the closed set K_n; therefore $x \in K_n$. If x, y both lie in every K_n, then $d(x, y) \le d(x, x_n) + d(x_n, y) \le 2r_n$ for every n, hence $x = y$. □

Theorem 1.4 (Baire category theorem)**.** *Every complete metric space is a Baire space (Definition 1.2). Equivalently, if $U_1, U_2, \dots$ are open dense subsets of a complete metric space X, then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.*

Proof. The assertion is immediate if X is empty. Otherwise, let V be any nonempty open set. Choose x_1 and $0 < r_1 \le 2^{-1}$ so that $\overline{B}(x_1, r_1) \subseteq V \cap U_1$. This is possible because $V \cap U_1$ is nonempty and open: choose an open ball inside it, then use a strictly smaller radius for the closed ball. Inductively choose

$$\overline{B}(x_{n+1}, r_{n+1}) \subseteq B(x_n, r_n) \cap U_{n+1}, \qquad 0 < r_{n+1} \le 2^{-(n+1)}.$$

Lemma 1.3 gives a point in all these closed balls. It belongs to V and every U_n. Since V was arbitrary, the intersection is dense. □

Corollary 1.5 (Closed-cover form of Baire)**.** *If a nonempty complete metric space X is covered by countably many closed sets F_n, then at least one F_n has nonempty interior.*

Proof. If every F_n had empty interior, each complement $X \setminus F_n$ would be open and dense. Theorem 1.4 would make their intersection dense and therefore nonempty, contradicting $X = \bigcup_n F_n$. □

Corollary 1.6 (Countable intersections of residual sets)**.** *In a complete metric space X, every countable intersection of residual sets is residual and dense, with "residual" as in Definition 1.2.*

Proof draft — unfinished. Write $X \setminus R_j = \bigcup_{k=1}^{\infty} A_{j,k}$, where each $A_{j,k}$ is nowhere dense. Then

$$X \setminus \bigcap_{j=1}^{\infty} R_j = \bigcup_{j,k \geq 1} A_{j,k},$$

so the intersection is residual because $\mathbb{N} \times \mathbb{N}$ is countable.

Still to write: replace the $A_{j,k}$ by their closures, take complements, and use Theorem 1.4 to justify density. Residual sets themselves need not be open, so applying Baire directly to the R_j would leave a gap.

Corollary 1.7 (Complete spaces without isolated points)**.** *A nonempty complete metric space with no isolated points is uncountable.*

Proof. If $X = \{x_1, x_2, \ldots\}$, the closed sets $F_n = \{x_n\}$ cover X. None has interior, since a singleton is open precisely when its point is isolated. This contradicts Corollary 1.5. The same argument covers a finite enumeration by repetition. □

2 Banach spaces and linear operators

Definition 2.1 (Normed and Banach spaces)**.** Let $\mathbb{K}$ be either $\mathbb{R}$ or $\mathbb{C}$. A norm on a $\mathbb{K}$-vector space X is a function $\|\cdot\| : X \to [0, \infty)$ such that $\|x\| = 0$ exactly when $x = 0$, $\|\lambda x\| = |\lambda| \|x\|$, and $\|x + y\| \leq \|x\| + \|y\|$. A normed space is *Banach* if the metric $d(x, y) = \|x - y\|$ is complete (Definition 1.1). For a normed space X, write $B_X(0, r) = \{x \in X : \|x\| < r\}$.

Lemma 2.2 (Closed subspaces are Banach)**.** *Every closed linear subspace M of a Banach space X is Banach under the restricted norm from Definition 2.1.*

Proof. A Cauchy sequence in M is also Cauchy in X, so it converges to some $x \in X$. Closedness gives $x \in M$, and convergence uses the same norm. □

Lemma 2.3 (Products of Banach spaces)**.** *If X and Y are Banach spaces (Definition 2.1), then $X \times Y$ is Banach for the norm $\|(x, y)\| = \|x\|_X + \|y\|_Y$.*

Proof. The norm axioms follow by applying those in X and Y separately. If $((x_n, y_n))$ is Cauchy, both coordinate sequences are Cauchy. Let their limits be $x \in X$ and $y \in Y$. Then $\|(x_n, y_n) - (x, y)\| = \|x_n - x\| + \|y_n - y\| \to 0$. □

Definition 2.4 (Bounded linear operators)**.** For normed spaces over the same field (Definition 2.1), a linear map $T : X \to Y$ is *bounded* if there is $C \geq 0$ such that $\|Tx\| \leq C\|x\|$ for every $x \in X$. Write $\mathcal{L}(X, Y)$ for these operators and set

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\|.$$

This is the least possible bound C. In particular, $\|Tx\| \leq \|T\|\|x\|$ for all x.

Lemma 2.5 (Boundedness and continuity)**.** *A linear map between normed spaces is continuous if and only if it is bounded as in Definition 2.4.*

Proof. A bounded map satisfies $\|Tx - Ty\| \leq \|T\|\|x - y\|$, so it is continuous. Conversely, continuity at zero gives $\delta > 0$ such that $\|z\| < \delta$ implies $\|Tz\| < 1$. For $x \neq 0$, take $z = \delta x/(2\|x\|)$. Linearity gives $\|Tx\| < 2\|x\|/\delta$; for $x = 0$ the bound is automatic. Thus T is bounded. □

Lemma 2.6 (Absolutely summable series)**.** *In a Banach space X (Definition 2.1), if $\sum_{n=1}^{\infty} \|x_n\| < \infty$, then the partial sums of $\sum_{n=1}^{\infty} x_n$ converge to an element $x \in X$, and $\|x\| \leq \sum_{n=1}^{\infty} \|x_n\|$.*

Proof. For $m > n$, the distance between the partial sums is at most $\sum_{k=n+1}^{m} \|x_k\|$, which tends to zero uniformly in $m > n$. Completeness gives a limit x. Passing to the limit in $\|\sum_{k=1}^{n} x_k\| \leq \sum_{k=1}^{n} \|x_k\|$ gives the estimate. □

Theorem 2.7 (Finite-dimensional subspaces are closed)**.** *Every finite-dimensional linear subspace of a normed space (Definition 2.1) is closed.*

Established result; proof not yet included. This is standard finite-dimensional normed-space theory, used for example in [3, solution to Exercise 2.4]. The ambient space need not be complete.

Lemma 2.8 (Proper subspaces have empty interior)**.** *Every proper linear subspace M of a normed space X (Definition 2.1) has empty interior.*

Proof. If $B_X(x_0, r) \subseteq M$, then $x_0 \in M$, and subtracting x_0 gives $B_X(0, r) \subseteq M$. For any nonzero $x \in X$, a sufficiently small positive multiple tx lies in this ball. Since M is a linear subspace, $x = t^{-1}(tx)$ also lies in M. This forces $M = X$. □

3 Uniform boundedness and convergence

Theorem 3.1 (Uniform boundedness principle)**.** *Let X be Banach, let Y be normed, and let $\mathcal{T} \subseteq \mathcal{L}(X, Y)$ be a nonempty family of bounded linear operators (Definition 2.4). If*

$$\sup_{T \in \mathcal{T}} \|Tx\| < \infty \qquad \text{for every } x \in X,$$

then $\sup_{T \in \mathcal{T}} \|T\| < \infty$. The family need not be countable, and Y need not be complete.

Proof. For $m \geq 1$, set $F_m = \{x \in X : \|Tx\| \leq m \text{ for every } T \in \mathcal{T}\}$. Each F_m is an intersection of closed sets, since every T is continuous by Lemma 2.5. The hypothesis gives $X = \bigcup_{m \geq 1} F_m$.

By Corollary 1.5, some F_m contains a ball $B_X(x_0, r)$. For $\|u\| \le 1$, both x_0 and $x_0 + (r/2)u$ belong to F_m, so, for every $T \in \mathcal{T}$,

$$\frac{r}{2}\|Tu\| \le \|T(x_0 + (r/2)u)\| + \|Tx_0\| \le 2m.$$

Taking the supremum over the unit ball gives $\|T\| \le 4m/r$, independently of T. □

Corollary 3.2 (Pointwise limits of operators)**.** *Let X be Banach and Y normed. Suppose $T_n \in \mathcal{L}(X, Y)$ and $T_n x$ converges in Y for every $x \in X$. Then $Tx = \lim_{n\to\infty} T_n x$ defines an operator $T \in \mathcal{L}(X, Y)$, and $M = \sup_n \|T_n\| < \infty$ with $\|T\| \le M$.*

Proof draft — unfinished. For each fixed x, the convergent sequence $(T_n x)$ is bounded. Theorem 3.1 therefore gives a common bound $M = \sup_n \|T_n\| < \infty$.

Still to write: pass to the limit in $T_n(ax + by) = aT_n x + bT_n y$ and in $\|T_n x\| \le M\|x\|$. These two steps must establish linearity and boundedness in the sense of Definition 2.4. No completeness of Y is needed because the limits in Y are part of the hypothesis.

Theorem 3.3 (Convergence from a dense subspace)**.** *Let X, Y be Banach spaces and $D \subseteq X$ a dense linear subspace. Suppose $T_n \in \mathcal{L}(X, Y)$, $M = \sup_n \|T_n\| < \infty$, and $(T_n d)$ converges in Y for each $d \in D$. Then $T_n x$ converges for every $x \in X$, and the limit is a bounded linear operator $T : X \to Y$ with $\|T\| \le M$.*

Established consequence; proof to be written. First use density and the operator estimate in Definition 2.4 to make $(T_n x)$ Cauchy. Completeness of Y from Definition 2.1 supplies the limit; then use Corollary 3.2. This is a standard density argument, not an unresolved assertion about Banach spaces.

Corollary 3.4 (Generic unboundedness)**.** *Let X be Banach, Y normed, and $\mathcal{T} \subseteq \mathcal{L}(X, Y)$ a nonempty family with $\sup_{T\in\mathcal{T}} \|T\| = \infty$. Then*

$$R = \{x \in X : \sup_{T\in\mathcal{T}} \|Tx\| = \infty\}$$

is a dense residual subset of X, in the sense of Definition 1.2.

Proof. Use the closed sets F_m from the proof of Theorem 3.1; their closedness does not require pointwise boundedness. If any F_m contained a ball, the same translated ball estimate would give a uniform bound on all operator norms, contrary to the hypothesis. Thus every F_m is closed with empty interior. Since $X \setminus R = \bigcup_m F_m$, R is residual. Also $R = \bigcap_m (X \setminus F_m)$ is dense by Theorem 1.4. □

4 Open mapping and its consequences

Lemma 4.1 (Approximate openness)**.** *Let X be normed, let Y be Banach, and let $T \in \mathcal{L}(X, Y)$ be surjective. There is $\delta > 0$ such that*

$$B_Y(0, \delta) \subseteq \overline{T(B_X(0, 1))}.$$

The overline here means closure in Y, not the image of a closed ball.

Proof. Surjectivity implies $Y = \bigcup_{n\ge1} T(B_X(0, n))$ and hence $Y = \bigcup_{n\ge1} \overline{T(B_X(0, n))}$. Corollary 1.5 gives $N \ge 1$, $y_0 \in Y$, and $r > 0$ with $B_Y(y_0, r) \subseteq \overline{T(B_X(0, N))}$. For $\|y\| < r$, both

$y_0 + y$ and y_0 belong to this closure. Choose $a_k, b_k \in B_X(0,N)$ with $Ta_k \to y_0 + y$ and $Tb_k \to y_0$. Then $T(a_k - b_k) \to y$ and $\|a_k - b_k\| < 2N$, so $B_Y(0,r) \subseteq \overline{T(B_X(0,2N))}$. Scaling by $1/(2N)$ gives the assertion with $\delta = r/(2N)$. The notation for balls and bounded operators is from Definitions 2.1 and 2.4. □

Lemma 4.2 (Geometric-series lifting)**.** *Let X be Banach, Y normed, and $T \in \mathcal{L}(X,Y)$. If $B_Y(0,\delta) \subseteq \overline{T(B_X(0,1))}$ for some $\delta > 0$, then*

$$B_Y(0,\delta/2) \subseteq T(B_X(0,1)).$$

Proof. Scaling the assumed inclusion gives $B_Y(0,\delta 2^{-n}) \subseteq \overline{T(B_X(0,2^{-n}))}$ for $n \geq 1$. Fix $\|y\| < \delta/2$. Recursively choose x_n with

$$\|x_n\| < 2^{-n}, \qquad \left\| y - T\sum_{k=1}^{n} x_k \right\| < \delta 2^{-(n+1)}.$$

For $n = 1$ the initial residual is y; at each later step the preceding residual lies in $B_Y(0,\delta 2^{-n})$, so the scaled closure inclusion allows an approximation with the displayed accuracy. Lemma 2.6 gives $x = \sum_{n\geq 1} x_n \in X$ and

$$\|x\| \leq \sum_{n\geq 1} \|x_n\| \leq \|x_1\| + \sum_{n\geq 2} 2^{-n} < \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

Continuity of T (Lemma 2.5) and convergence of the residuals give $Tx = y$. Notice that this step uses completeness of X; it does not require completeness of Y. □

Theorem 4.3 (Open mapping theorem)**.** *Every surjective bounded linear operator $T : X \to Y$ between Banach spaces is open: $T(U)$ is open in Y whenever U is open in X.*

Proof. Lemma 4.1 gives $\delta > 0$ with a ball contained in the closure of the image of the unit ball. Lemma 4.2 removes that closure and gives $B_Y(0,\delta/2) \subseteq T(B_X(0,1))$. Now let $x \in U$ and choose $\varepsilon > 0$ with $x + B_X(0,\varepsilon) \subseteq U$. By linearity and scaling,

$$Tx + B_Y(0,\varepsilon\delta/2) \subseteq T(x + B_X(0,\varepsilon)) \subseteq T(U).$$

Every point of $T(U)$ is therefore an interior point. □

Corollary 4.4 (Bounded inverse theorem)**.** *A bijective bounded linear operator between Banach spaces has a bounded linear inverse.*

Proof. If $T : X \to Y$ is such an operator, elementary linear algebra makes T^{-1} linear. For every open $U \subseteq X$, $(T^{-1})^{-1}(U) = T(U)$ is open by Theorem 4.3. Thus T^{-1} is continuous, and Lemma 2.5 makes it bounded. □

Corollary 4.5 (Equivalence of comparable complete norms)**.** *Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be two norms on the same vector space V, both making V Banach. If there is $C > 0$ such that $\|x\|_b \leq C\|x\|_a$ for all $x \in V$, then there is $D > 0$ with $\|x\|_a \leq D\|x\|_b$ for all $x \in V$. Thus the norms are equivalent.*

Established corollary; proof not yet included. Apply Corollary 4.4 to the identity map from $(V, \|\cdot\|_a)$ onto $(V, \|\cdot\|_b)$. Both completeness assumptions matter in this argument; compare [2, Corollary 8].

5 Closed graphs

Definition 5.1 (Graph of an operator)**.** For normed spaces X, Y and an everywhere-defined linear map $T : X \to Y$, its graph is the linear subspace $G(T) = \{(x, Tx) : x \in X\} \subseteq X \times Y$. The map has a *closed graph* if $G(T)$ is closed in the sum norm $\|(x, y)\| = \|x\| + \|y\|$ used in Lemma 2.3. Equivalently, $x_n \to x$ and $Tx_n \to y$ imply $y = Tx$. Closedness does not, by definition, assert that Tx_n converges whenever x_n does.

Theorem 5.2 (Closed graph theorem)**.** *Let X and Y be Banach spaces and let $T : X \to Y$ be an everywhere-defined linear map. If its graph is closed in the sense of Definition 5.1, then T is bounded.*

Proof draft — unfinished. By Lemma 2.3, $X \times Y$ is Banach. The graph $G(T)$ is a closed linear subspace, so Lemma 2.2 makes it Banach in the restricted sum norm. The projection $P : G(T) \to X$, $P(x, Tx) = x$, is linear, bijective, and bounded because $\|x\| \le \|x\| + \|Tx\|$. Corollary 4.4 gives a bounded inverse $P^{-1} : X \to G(T)$.

Still to write: introduce the second-coordinate projection $Q : G(T) \to Y$, prove its norm estimate, and use $T = QP^{-1}$ to obtain an explicit bound for $\|Tx\|$ in terms of $\|x\|$. The boundedness of Q must come from the product norm, not from assuming the conclusion that T is already bounded.

6 A question about dimension

Conjecture 6.1 (No countable Hamel basis)**.** *An infinite-dimensional Banach space has no countable Hamel basis. Here a Hamel basis is a linearly independent set such that every vector is a* finite *linear combination of basis vectors.*

Notebook status: conjecture, not proved here. This is a classical exercise, recorded as an unresolved task in these working notes; it is not open in the mathematical literature. If a basis could be enumerated as (e_n), the candidate sets to examine would be $F_n = \operatorname{span}\{e_1, \dots, e_n\}$. Try combining Theorem 2.7, Lemma 2.8, and Corollary 1.5. Be careful: finite linear combinations are different from convergent infinite series. Separability, meaning existence of a countable dense subset, does not assert existence of a countable Hamel basis.

References

[1] Christoph Kehle, *Introduction to Functional Analysis*, MIT 18.102, Spring 2025, Sections 1.5 and 2.3. Course notes.

[2] Orr Shalit, *Advanced Analysis, Notes 9: Banach spaces (the three big theorems)*, 2012. Author's lecture notes.

[3] Alexander C. R. Belton, *Functional Analysis*, revised edition, 2006, Chapters 1–2 and the solutions to exercises. Lecture notes.